

1 INTRODUCTION

Agitators play a pivotal role in facilitating the homogeneous mixing of substances, a fundamental process in pharmaceutical manufacturing. OQ, in this context, serves as a comprehensive evaluation and verification process, encompassing various key parameters such as agitation speed, mixing direction, intensity, load handling capabilities, temperature control, safety features, noise and vibration levels, material compatibility, operational reliability, and documentation.

1.1 ABBREVIATIONS

OQ - Operational Qualification
SOP - Standard Operating Procedure
PFM - Revolutions per Minute
HVAC - Heating, Ventilation, and Air Conditioning
HMI - Human-Machine Interface
PLC - Programmable Logic Controller

2 PURPOSE

The purpose of OQ is to rigorously assess the agitator's ability to function within specified parameters, ensuring it meets not only process requirements but also stringent regulatory and safety standards. A successful OQ not only validates the agitator's performance but also enhances process consistency, product quality, and overall operational excellence in the chemical processing industry. The OQ Protocol section of this qualification package defines and validates that the Agitator will perform its operational functions repeatedly, reliably, and as expected throughout the life of the system. The OQ Protocol documentation includes confirmation reports of operational functionality tests used to verify that the Agitator will perform its operational functions repeatedly, reliably, and as expected throughout the life of the system. The OQ Protocol documentation includes quality inspection reports, and checklists of the Equipment System labeling and documentation. The OQ Protocol documentation also includes the Operating and Maintenance Manual.

3 SYSTEM DESCRIPTION

Equipment : Agitator
Function : Agitation/ Mixing of Liquid Samples
Year of manufacture : 2022-2023

(Equipment Id. _____)

4 SCOPE

This protocol applies to the operational qualification of Agitator

4.1 PROTOCOL ACCEPTANCE CRITERIA

- All required test steps have been performed and actual results meet the expected results.
- All corresponding attachments are completed.
- All tests are passed and/or exceptional conditions have been adequately resolved and approved.



Dynamic Agitator



Magnetic Agitator

5 PRE-REQUISITES

Prior to executing qualification, following condition must be fulfilled:

Sr. No.	Pre-requisites	Availability (Yes / No / NA)	Document no. & location
1.	SAT protocol of equipment is pre-approved.		
2.	Equipment is installed and tagged.		
3.	Vessel is empty, clean, and not pressurized.		
4.	All utilities are available and connected to system.		
5.	All utilities are available as per Pre SAT-Check list		

5 OPERATIONAL QUALIFICATION SECTION:

Operational Qualification Test Criteria 1

☐ Mark if not applicable

No.	Instructions	Reference	Expected Results	Actual Result	Result (Pass/Fail)	Signature
OQ_1	Face Velocity Test: Turn on main power, turn on control box power, set speed to deliver 72-108 fpm. Measure airflow using calibrated device	FRS_FS_01	Equipment powers up, fans start and delivers 72-108 fpm.			
OQ_2	Static Particulate Test: Measure room particulates at the work surface using a calibrated meter. Measure in grid within tent work area (minimum 8 sample points)	FRS_FS_02	Particulate count less than 10,000 - 0.5-micron particles.			
OQ_3	Pressurization: Measure the differential pressure from the inside of the tent to the outside using a calibrated meter	FRS_FS_03	Differential pressure is positive from inside of tent to outside			
OQ_4	Room Change Rate: Calculate tent room change rates. (512 cu.ft. room volume; design airflow 700 cfm per filter)	FRS_MBS_01	Verify 60 changes per hour			
OQ_5	Manually operate the control switches listed in the applicable attachment	FRS_MBS_02	All operator controls and indicators tested shall conform to the expected response			

No.	Instructions	Reference	Expected Results	Actual Result	Result (Pass/Fail)	Signature
OQ_6	Confirmation of Receipt of all Required Parts: 1. Review manufacturing drawing, bill-of-materials, and sales order 2. Confirm that all loose plumbing and/or electrical components were shipped	FRS_MBS_03	All plumbing and/or electrical components were shipped			
OQ_7	Check whether the Equipment maintains temperature of - 70°C for 7 days	FRS_DS_01	Maintains temperature over 7 days			
OQ_8	Check whether temperature LED functions correctly	FRS_DS_02	LED temperature display functions correctly			
OQ_9	Confirmation of Components are Leak Tight after Connection to LN2 Supply: 1. Review safe handling of liquid nitrogen guidelines 2. Hook up liquid nitrogen supply to T-W Equipment per Operating Manual 3. Open fill valve on liquid nitrogen supply 4. Inspect component connections for leaks using leak test procedures 5. Tighten any leaks, if required	FRS_DS_03	Components are leak tight after connection to LN2 supply			

No.	Instructions	Reference	Expected Results	Actual Result	Result (Pass/Fail)	Signature
OQ_10	Confirmation of High Temperature Alarm Set Point: Confirm whether Alarm rings if Temperature rises more than the optimum level	FRS_DS_04	Alarm Rings once temperature rises more than the optimum level			
OQ_11	Confirmation of Low Temperature Alarm Set Point: Confirm whether Alarm rings if Temperature drops below the optimum level	FRS_DS_05	Alarm Rings once temperature drops below the optimum level			

Operational Qualification Test Criteria 2

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No.	Instructions	Reference	Expected Results	Actual Result	Result (Pass/Fail)	Signature
IQ_12	Verification should be done to confirm whether erosion resistant coatings based on the design of the turbine or pump and the severity of erosion are used	FRS_FS_04	Verification for erosion resistant coatings is successful			
IQ_13	Equipment verification done to ensure that the machine has maintained GMP conditions at all times with no possibility for product mix or for rejected product to be released.	FRS_FS_05	Equipment Verification for GMP maintenance is successful			

No.	Instructions	Reference	Expected Results	Actual Result	Result (Pass/Fail)	Signature
IQ_14	Verification is done to confirm environment setting is as described in the ICD (Interface Control Documents)	FRS_IS_01	Environment setting as per ICD (Interface Control Documents) is verified successfully			
IQ_15	Verification to be done to confirm whether equipment complies with the ISS System Services interface	FRS_IS_02	Equipment compilation with the ISS System Services interface is verified successfully			

Operational Qualification Test Criteria 3

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No.	Instructions	Reference	Expected Results	Actual Result	Result (Pass/Fail)	Signature
OQ_16	Safety Features: Confirm Agitator includes functional safety interlocks and emergency stop	FRS_PR_11	Functional safety interlocks and emergency stop operate			
OQ_17	Safety Features: Confirm agitator safely shuts down in case of anomalies or faults	FRS_PR_12	Agitator safely shuts down in case of anomalies			
OQ_18	Noise and Vibration: Measure and document noise and vibration levels	FRS_PR_13	Noise and vibration levels comply with standards			

No.	Instructions	Reference	Expected Results	Actual Result	Result (Pass/Fail)	Signature
OQ_19	Noise and Vibration: Confirm compliance of equipment with safety and environmental standards	FRS_PR_14	Equipment is compliant with safety and environmental standards			
OQ_20	Material Compatibility: Confirm material compatibility with various processed materials	FRS_PR_15	Agitator is compatible with various processed materials			
OQ_21	Material Compatibility: Verify if there are any material specific issues (Eg. Corrosion)	FRS_PR_16	No material specific issues are observed			
OQ_22	Operational Reliability: Verify continuous operation of agitator to assess long-term reliability	FRS_PR_17	Agitator demonstrates long-term reliability			
OQ_23	Operational Reliability: Monitor for signs of wear, overheating etc. during continuous operation	FRS_PR_18	No signs of wear, overheating or similar issues are observed			
OQ_24	Documentation: Verify thorough documentation of parameters and test results	FRS_PR_19	Thorough documentation of parameters and results			

No.	Instructions	Reference	Expected Results	Actual Result	Result (Pass/Fail)	Signature
OQ_25	Documentation: Compliance with regulatory and quality standards	FRS_PR_20	Equipment is compliant with regulatory and quality standards			

Operational Qualification Test Criteria 4

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No.	Instructions	Reference	Expected Results	Actual Result	Specification Met?	Signature	Comments
OQ_26	Agitation Speed: Verify specified speed range within acceptable tolerance of +/- 5%	FRS_PR_01	Equipment speed is with acceptable tolerance of +/- 5%		Met Not Met		
OQ_27	Agitation Speed: Confirm that the agitator operates at consistent speed within tolerances.	FRS_PR_02	Equipment operates at consistent speed within tolerances.		Met Not Met		
OQ_28	Mixing direction: Confirm that the agitator startup & stop is smooth	FRS_PR_03	Smooth startup & stop functionality is observed		Met Not Met		
OQ_29	Mixing direction: Test the clockwise(CW) and counter clockwise(CCW) operation of agitator	FRS_PR_04	Agitator changes direction(CW and CCW) as required		Met Not Met		

No.	Instructions	Reference	Expected Results	Actual Result	Specification Met?	Signature	Comments
OQ_30	Mixing Intensity: Confirm smooth direction changes of agitator from CW and CCW	FRS_PR_05	Agitator direction change occurs smoothly, without jolts.		Met Not Met		
OQ_31	Mixing Intensity: Confirm that provides required mixing intensity ensuring effective mixing of components.	FRS_PR_06	Agitator provides required mixing intensity		Met Not Met		
OQ_32	Confirm that Mixing Intensity levels are adjustable within specified parameters	FRS_PR_07	Agitator Mixing Intensity is adjustable within parameters		Met Not Met		
OQ_33	Load Handling: Confirm that Agitator performance under varying load conditions within design capacity	FRS_PR_08	Agitator performs under varying load conditions		Met Not Met		
OQ_34	Temperature Control: Confirm that agitator maintains desired vessel temperature during mixing	FRS_PR_09	Agitator maintains desired vessel temperature		Met Not Met		
OQ_35	Temperature Control: Agitator responds to temperature set point changes	FRS_PR_10	Agitator prompts response to temperature set point changes		Met Not Met		

Operational Qualification Test Criteria 5

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S.No	Test Description			Adheres to Specification (Yes/No)
OQ_36	Agitation Speed Parameter Verification:			
	S. No	Test description	Achieves (Yes/No)	
	1	Set 10 rpm as the target speed for agitator using the control knobs on the equipment control panel		
	2	Set 50 rpm as the target speed for agitator using the control knobs on the equipment control panel		
	3	Set 100 rpm as the target speed for agitator using the control knobs on the equipment control panel		
OQ_37.a	Pump Accuracy Verification (Delay Parameters):			
	S. No	Test description	Ref. Value	
	1	Delay Time (in milliseconds)	1-3 ms	
	2	Delay Volume (millilitres)	50-75 ml	
OQ_37.b	Pump Accuracy Verification (Mixing Parameters)			
	S. No	Test description	Ref. Value	
	1	Delay Time (in milliseconds)	1-3 ms	
	2	Delay Volume (millilitres)	50-75 ml	

6 SUMMARIZE OPERATIONAL QUALIFICATION TEST CRITERIA:

Description	Results
The equipment is operating in accordance with the manufacturer specifications on restoring power to the equipment.	☐ Yes ☐ No ☐ Exp.No. _____
Verified By:	Date:

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OQ_28	Mixing direction: Confirm that the agitator startup & stop is smooth	FRS_PR_03	Smooth startup & stop functionality is observed		Met Not Met		
OQ_29	Mixing direction: Test the clockwise(CW) and counter clockwise(CCW) operation of agitator	FRS_PR_04	Agitator changes direction(CW and CCW) as required		Met Not Met		

No.	Instructions	Reference	Expected Results	Actual Result	Specification Met?	Signature	Comments
OQ_30	Mixing Intensity: Confirm smooth direction changes of agitator from CW and CCW	FRS_PR_05	Agitator direction change occurs smoothly, without jolts.		Met Not Met		
OQ_31	Mixing Intensity: Confirm that provides required mixing intensity ensuring effective mixing of components.	FRS_PR_06	Agitator provides required mixing intensity		Met Not Met		
OQ_32	Confirm that Mixing Intensity levels are adjustable within specified parameters	FRS_PR_07	Agitator Mixing Intensity is adjustable within parameters		Met Not Met		
OQ_33	Load Handling: Confirm that Agitator performance under varying load conditions within design capacity	FRS_PR_08	Agitator performs under varying load conditions		Met Not Met		
OQ_34	Temperature Control: Confirm that agitator maintains desired vessel temperature during mixing	FRS_PR_09	Agitator maintains desired vessel temperature		Met Not Met		
OQ_35	Temperature Control: Agitator responds to temperature set point changes	FRS_PR_10	Agitator prompts response to temperature set point changes		Met Not Met		

Operational Qualification Test Criteria 5

☐ Mark if not applicable

S.No	Test Description			Adheres to Specification (Yes/No)
OQ_36	Agitation Speed Parameter Verification:			
	S. No	Test description	Achieves (Yes/No)	
	1	Set 10 rpm as the target speed for agitator using the control knobs on the equipment control panel		
	2	Set 50 rpm as the target speed for agitator using the control knobs on the equipment control panel		
	3	Set 100 rpm as the target speed for agitator using the control knobs on the equipment control panel		
OQ_37.a	Pump Accuracy Verification (Delay Parameters):			
	S. No	Test description	Ref. Value	
	1	Delay Time (in milliseconds)	1-3 ms	
	2	Delay Volume (millilitres)	50-75 ml	
OQ_37.b	Pump Accuracy Verification (Mixing Parameters)			
	S. No	Test description	Ref. Value	
	1	Delay Time (in milliseconds)	1-3 ms	
	2	Delay Volume (millilitres)	50-75 ml	

6 SUMMARIZE OPERATIONAL QUALIFICATION TEST CRITERIA:

Description	Results
The equipment is operating in accordance with the manufacturer specifications on restoring power to the equipment.	☐ Yes ☐ No ☐ Exp.No. _____
Verified By:	Date:

1 INTRODUCTION

Agitators play a pivotal role in facilitating the homogeneous mixing of substances, a fundamental process in pharmaceutical manufacturing. OQ, in this context, serves as a comprehensive evaluation and verification process, encompassing various key parameters such as agitation speed, mixing direction, intensity, load handling capabilities, temperature control, safety features, noise and vibration levels, material compatibility, operational reliability, and documentation.

1.1 ABBREVIATIONS

OQ - Operational Qualification
SOP - Standard Operating Procedure
PFM - Revolutions per Minute
HVAC - Heating, Ventilation, and Air Conditioning
HMI - Human-Machine Interface
PLC - Programmable Logic Controller

2 PURPOSE

The purpose of OQ is to rigorously assess the agitator's ability to function within specified parameters, ensuring it meets not only process requirements but also stringent regulatory and safety standards. A successful OQ not only validates the agitator's performance but also enhances process consistency, product quality, and overall operational excellence in the chemical processing industry. The OQ Protocol section of this qualification package defines and validates that the Agitator will perform its operational functions repeatedly, reliably, and as expected throughout the life of the system. The OQ Protocol documentation includes confirmation reports of operational functionality tests used to verify that the Agitator will perform its operational functions repeatedly, reliably, and as expected throughout the life of the system. The OQ Protocol documentation includes quality inspection reports, and checklists of the Equipment System labeling and documentation. The OQ Protocol documentation also includes the Operating and Maintenance Manual.

3 SYSTEM DESCRIPTION

Equipment : Agitator
Function : Agitation/ Mixing of Liquid Samples
Year of manufacture : 2022-2023

(Equipment Id. _____)

4 SCOPE

This protocol applies to the operational qualification of Agitator

4.1 PROTOCOL ACCEPTANCE CRITERIA

- All required test steps have been performed and actual results meet the expected results.
- All corresponding attachments are completed.
- All tests are passed and/or exceptional conditions have been adequately resolved and approved.



Dynamic Agitator



Magnetic Agitator

5 PRE-REQUISITES

Prior to executing qualification, following condition must be fulfilled:

Sr. No.	Pre-requisites	Availability (Yes / No / NA)	Document no. & location
1.	SAT protocol of equipment is pre-approved.		
2.	Equipment is installed and tagged.		
3.	Vessel is empty, clean, and not pressurized.		
4.	All utilities are available and connected to system.		
5.	All utilities are available as per Pre SAT-Check list		

5 OPERATIONAL QUALIFICATION SECTION:

Operational Qualification Test Criteria 1

☐ Mark if not applicable

No.	Instructions	Reference	Expected Results	Actual Result	Result (Pass/Fail)	Signature
OQ_1	Face Velocity Test: Turn on main power, turn on control box power, set speed to deliver 72-108 fpm. Measure airflow using calibrated device	FRS_FS_01	Equipment powers up, fans start and delivers 72-108 fpm.			
OQ_2	Static Particulate Test: Measure room particulates at the work surface using a calibrated meter. Measure in grid within tent work area (minimum 8 sample points)	FRS_FS_02	Particulate count less than 10,000 - 0.5-micron particles.			
OQ_3	Pressurization: Measure the differential pressure from the inside of the tent to the outside using a calibrated meter	FRS_FS_03	Differential pressure is positive from inside of tent to outside			
OQ_4	Room Change Rate: Calculate tent room change rates. (512 cu.ft. room volume; design airflow 700 cfm per filter)	FRS_MBS_01	Verify 60 changes per hour			
OQ_5	Manually operate the control switches listed in the applicable attachment	FRS_MBS_02	All operator controls and indicators tested shall conform to the expected response			

No.	Instructions	Reference	Expected Results	Actual Result	Result (Pass/Fail)	Signature
OQ_6	Confirmation of Receipt of all Required Parts: 1. Review manufacturing drawing, bill-of-materials, and sales order 2. Confirm that all loose plumbing and/or electrical components were shipped	FRS_MBS_03	All plumbing and/or electrical components were shipped			
OQ_7	Check whether the Equipment maintains temperature of -70°C for 7 days	FRS_DS_01	Maintains temperature over 7 days			
OQ_8	Check whether temperature LED functions correctly	FRS_DS_02	LED temperature display functions correctly			
OQ_9	Confirmation of Components are Leak Tight after Connection to LN2 Supply: 1. Review safe handling of liquid nitrogen guidelines 2. Hook up liquid nitrogen supply to T-W Equipment per Operating Manual 3. Open fill valve on liquid nitrogen supply 4. Inspect component connections for leaks using leak test procedures 5. Tighten any leaks, if required	FRS_DS_03	Components are leak tight after connection to LN2 supply			

No.	Instructions	Reference	Expected Results	Actual Result	Result (Pass/Fail)	Signature
OQ_10	Confirmation of High Temperature Alarm Set Point: Confirm whether Alarm rings if Temperature rises more than the optimum level	FRS_DS_04	Alarm Rings once temperature rises more than the optimum level			
OQ_11	Confirmation of Low Temperature Alarm Set Point: Confirm whether Alarm rings if Temperature drops below the optimum level	FRS_DS_05	Alarm Rings once temperature drops below the optimum level			

Operational Qualification Test Criteria 2

☐ Mark if not applicable

No.	Instructions	Reference	Expected Results	Actual Result	Result (Pass/Fail)	Signature
IQ_12	Verification should be done to confirm whether erosion resistant coatings based on the design of the turbine or pump and the severity of erosion are used	FRS_FS_04	Verification for erosion resistant coatings is successful			
IQ_13	Equipment verification done to ensure that the machine has maintained GMP conditions at all times with no possibility for product mix or for rejected product to be released.	FRS_FS_05	Equipment Verification for GMP maintenance is successful			

No.	Instructions	Reference	Expected Results	Actual Result	Result (Pass/Fail)	Signature
IQ_14	Verification is done to confirm environment setting is as described in the ICD (Interface Control Documents)	FRS_IS_01	Environment setting as per ICD (Interface Control Documents) is verified successfully			
IQ_15	Verification to be done to confirm whether equipment complies with the ISS System Services interface	FRS_IS_02	Equipment compilation with the ISS System Services interface is verified successfully			

Operational Qualification Test Criteria 3

☐ Mark if not applicable

No.	Instructions	Reference	Expected Results	Actual Result	Result (Pass/Fail)	Signature
OQ_16	Safety Features: Confirm Agitator includes functional safety interlocks and emergency stop	FRS_PR_11	Functional safety interlocks and emergency stop operate			
OQ_17	Safety Features: Confirm agitator safely shuts down in case of anomalies or faults	FRS_PR_12	Agitator safely shuts down in case of anomalies			
OQ_18	Noise and Vibration: Measure and document noise and vibration levels	FRS_PR_13	Noise and vibration levels comply with standards			

No.	Instructions	Reference	Expected Results	Actual Result	Result (Pass/Fail)	Signature
OQ_19	Noise and Vibration: Confirm compliance of equipment with safety and environmental standards	FRS_PR_14	Equipment is compliant with safety and environmental standards			
OQ_20	Material Compatibility: Confirm material compatibility with various processed materials	FRS_PR_15	Agitator is compatible with various processed materials			
OQ_21	Material Compatibility: Verify if there are any material specific issues (Eg. Corrosion)	FRS_PR_16	No material specific issues are observed			
OQ_22	Operational Reliability: Verify continuous operation of agitator to assess long-term reliability	FRS_PR_17	Agitator demonstrates long-term reliability			
OQ_23	Operational Reliability: Monitor for signs of wear, overheating etc. during continuous operation	FRS_PR_18	No signs of wear, overheating or similar issues are observed			
OQ_24	Documentation: Verify thorough documentation of parameters and test results	FRS_PR_19	Thorough documentation of parameters and results			

No.	Instructions	Reference	Expected Results	Actual Result	Result (Pass/Fail)	Signature
OQ_25	Documentation: Compliance with regulatory and quality standards	FRS_PR_20	Equipment is compliant with regulatory and quality standards			

Operational Qualification Test Criteria 4

☐ Mark if not applicable

No.	Instructions	Reference	Expected Results	Actual Result	Specification Met?	Signature	Comments
OQ_26	Agitation Speed: Verify specified speed range within acceptable tolerance of +/- 5%	FRS_PR_01	Equipment speed is with acceptable tolerance of +/- 5%		Met Not Met		
OQ_27	Agitation Speed: Confirm that the agitator operates at consistent speed within tolerances.	FRS_PR_02	Equipment operates at consistent speed within tolerances.		Met Not Met		
OQ_28	Mixing direction: Confirm that the agitator startup & stop is smooth	FRS_PR_03	Smooth startup & stop functionality is observed		Met Not Met		
OQ_29	Mixing direction: Test the clockwise(CW) and counter clockwise(CCW) operation of agitator	FRS_PR_04	Agitator changes direction(CW and CCW) as required		Met Not Met		

No.	Instructions	Reference	Expected Results	Actual Result	Specification Met?	Signature	Comments
OQ_30	Mixing Intensity: Confirm smooth direction changes of agitator from CW and CCW	FRS_PR_05	Agitator direction change occurs smoothly, without jolts.		Met Not Met		
OQ_31	Mixing Intensity: Confirm that provides required mixing intensity ensuring effective mixing of components.	FRS_PR_06	Agitator provides required mixing intensity		Met Not Met		
OQ_32	Confirm that Mixing Intensity levels are adjustable within specified parameters	FRS_PR_07	Agitator Mixing Intensity is adjustable within parameters		Met Not Met		
OQ_33	Load Handling: Confirm that Agitator performance under varying load conditions within design capacity	FRS_PR_08	Agitator performs under varying load conditions		Met Not Met		
OQ_34	Temperature Control: Confirm that agitator maintains desired vessel temperature during mixing	FRS_PR_09	Agitator maintains desired vessel temperature		Met Not Met		
OQ_35	Temperature Control: Agitator responds to temperature set point changes	FRS_PR_10	Agitator prompts response to temperature set point changes		Met Not Met		

Operational Qualification Test Criteria 5

☐ Mark if not applicable

S.No	Test Description			Adheres to Specification (Yes/No)
OQ_36	Agitation Speed Parameter Verification:			
	S. No	Test description	Achieves (Yes/No)	
	1	Set 10 rpm as the target speed for agitator using the control knobs on the equipment control panel		
	2	Set 50 rpm as the target speed for agitator using the control knobs on the equipment control panel		
	3	Set 100 rpm as the target speed for agitator using the control knobs on the equipment control panel		
OQ_37.a	Pump Accuracy Verification (Delay Parameters):			
	S. No	Test description	Ref. Value	
	1	Delay Time (in milliseconds)	1-3 ms	
	2	Delay Volume (millilitres)	50-75 ml	
OQ_37.b	Pump Accuracy Verification (Mixing Parameters)			
	S. No	Test description	Ref. Value	
	1	Delay Time (in milliseconds)	1-3 ms	
	2	Delay Volume (millilitres)	50-75 ml	

6 SUMMARIZE OPERATIONAL QUALIFICATION TEST CRITERIA:

Description	Results
The equipment is operating in accordance with the manufacturer specifications on restoring power to the equipment.	☐ Yes ☐ No ☐ Exp.No. _____
Verified By:	Date:

1 INTRODUCTION

Agitators play a pivotal role in facilitating the homogeneous mixing of substances, a fundamental process in pharmaceutical manufacturing. OQ, in this context, serves as a comprehensive evaluation and verification process, encompassing various key parameters such as agitation speed, mixing direction, intensity, load handling capabilities, temperature control, safety features, noise and vibration levels, material compatibility, operational reliability, and documentation.

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3 SYSTEM DESCRIPTION

Equipment : Agitator
Function : Agitation/ Mixing of Liquid Samples
Year of manufacture : 2022-2023

(Equipment Id. _____)

4 SCOPE

This protocol applies to the operational qualification of Agitator

4.1 PROTOCOL ACCEPTANCE CRITERIA

- All required test steps have been performed and actual results meet the expected results.
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Dynamic Agitator



Magnetic Agitator

5 PRE-REQUISITES

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Sr. No.	Pre-requisites	Availability (Yes / No / NA)	Document no. & location
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5.	All utilities are available as per Pre SAT-Check list		

5 OPERATIONAL QUALIFICATION SECTION:

Operational Qualification Test Criteria 1

☐ Mark if not applicable

No.	Instructions	Reference	Expected Results	Actual Result	Result (Pass/Fail)	Signature
OQ_1	Face Velocity Test: Turn on main power, turn on control box power, set speed to deliver 72-108 fpm. Measure airflow using calibrated device	FRS_FS_01	Equipment powers up, fans start and delivers 72-108 fpm.			
OQ_2	Static Particulate Test: Measure room particulates at the work surface using a calibrated meter. Measure in grid within tent work area (minimum 8 sample points)	FRS_FS_02	Particulate count less than 10,000 - 0.5-micron particles.			
OQ_3	Pressurization: Measure the differential pressure from the inside of the tent to the outside using a calibrated meter	FRS_FS_03	Differential pressure is positive from inside of tent to outside			
OQ_4	Room Change Rate: Calculate tent room change rates. (512 cu.ft. room volume; design airflow 700 cfm per filter)	FRS_MBS_01	Verify 60 changes per hour			
OQ_5	Manually operate the control switches listed in the applicable attachment	FRS_MBS_02	All operator controls and indicators tested shall conform to the expected response			

No.	Instructions	Reference	Expected Results	Actual Result	Result (Pass/Fail)	Signature
OQ_6	Confirmation of Receipt of all Required Parts: 1. Review manufacturing drawing, bill-of-materials, and sales order 2. Confirm that all loose plumbing and/or electrical components were shipped	FRS_MBS_03	All plumbing and/or electrical components were shipped			
OQ_7	Check whether the Equipment maintains temperature of -70°C for 7 days	FRS_DS_01	Maintains temperature over 7 days			
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No.	Instructions	Reference	Expected Results	Actual Result	Result (Pass/Fail)	Signature
OQ_10	Confirmation of High Temperature Alarm Set Point: Confirm whether Alarm rings if Temperature rises more than the optimum level	FRS_DS_04	Alarm Rings once temperature rises more than the optimum level			
OQ_11	Confirmation of Low Temperature Alarm Set Point: Confirm whether Alarm rings if Temperature drops below the optimum level	FRS_DS_05	Alarm Rings once temperature drops below the optimum level			

Operational Qualification Test Criteria 2

☐ Mark if not applicable

No.	Instructions	Reference	Expected Results	Actual Result	Result (Pass/Fail)	Signature
IQ_12	Verification should be done to confirm whether erosion resistant coatings based on the design of the turbine or pump and the severity of erosion are used	FRS_FS_04	Verification for erosion resistant coatings is successful			
IQ_13	Equipment verification done to ensure that the machine has maintained GMP conditions at all times with no possibility for product mix or for rejected product to be released.	FRS_FS_05	Equipment Verification for GMP maintenance is successful			

No.	Instructions	Reference	Expected Results	Actual Result	Result (Pass/Fail)	Signature
IQ_14	Verification is done to confirm environment setting is as described in the ICD (Interface Control Documents)	FRS_IS_01	Environment setting as per ICD (Interface Control Documents) is verified successfully			
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Operational Qualification Test Criteria 3

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No.	Instructions	Reference	Expected Results	Actual Result	Result (Pass/Fail)	Signature
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OQ_21	Material Compatibility: Verify if there are any material specific issues (Eg. Corrosion)	FRS_PR_16	No material specific issues are observed			
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OQ_24	Documentation: Verify thorough documentation of parameters and test results	FRS_PR_19	Thorough documentation of parameters and results			

No.	Instructions	Reference	Expected Results	Actual Result	Result (Pass/Fail)	Signature
OQ_25	Documentation: Compliance with regulatory and quality standards	FRS_PR_20	Equipment is compliant with regulatory and quality standards			

Operational Qualification Test Criteria 4

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No.	Instructions	Reference	Expected Results	Actual Result	Specification Met?	Signature	Comments
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OQ_28	Mixing direction: Confirm that the agitator startup & stop is smooth	FRS_PR_03	Smooth startup & stop functionality is observed		Met Not Met		
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4 SCOPE

This protocol applies to the operational qualification of Agitator

4.1 PROTOCOL ACCEPTANCE CRITERIA

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Dynamic Agitator



Magnetic Agitator

5 PRE-REQUISITES

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5 OPERATIONAL QUALIFICATION SECTION:

Operational Qualification Test Criteria 1

☐ Mark if not applicable

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OQ_5	Manually operate the control switches listed in the applicable attachment	FRS_MBS_02	All operator controls and indicators tested shall conform to the expected response			

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No.	Instructions	Reference	Expected Results	Actual Result	Result (Pass/Fail)	Signature
OQ_10	Confirmation of High Temperature Alarm Set Point: Confirm whether Alarm rings if Temperature rises more than the optimum level	FRS_DS_04	Alarm Rings once temperature rises more than the optimum level			
OQ_11	Confirmation of Low Temperature Alarm Set Point: Confirm whether Alarm rings if Temperature drops below the optimum level	FRS_DS_05	Alarm Rings once temperature drops below the optimum level			

Operational Qualification Test Criteria 2

☐ Mark if not applicable

No.	Instructions	Reference	Expected Results	Actual Result	Result (Pass/Fail)	Signature
IQ_12	Verification should be done to confirm whether erosion resistant coatings based on the design of the turbine or pump and the severity of erosion are used	FRS_FS_04	Verification for erosion resistant coatings is successful			
IQ_13	Equipment verification done to ensure that the machine has maintained GMP conditions at all times with no possibility for product mix or for rejected product to be released.	FRS_FS_05	Equipment Verification for GMP maintenance is successful			

No.	Instructions	Reference	Expected Results	Actual Result	Result (Pass/Fail)	Signature
IQ_14	Verification is done to confirm environment setting is as described in the ICD (Interface Control Documents)	FRS_IS_01	Environment setting as per ICD (Interface Control Documents) is verified successfully			
IQ_15	Verification to be done to confirm whether equipment complies with the ISS System Services interface	FRS_IS_02	Equipment compilation with the ISS System Services interface is verified successfully			

Operational Qualification Test Criteria 3

☐ Mark if not applicable

No.	Instructions	Reference	Expected Results	Actual Result	Result (Pass/Fail)	Signature
OQ_16	Safety Features: Confirm Agitator includes functional safety interlocks and emergency stop	FRS_PR_11	Functional safety interlocks and emergency stop operate			
OQ_17	Safety Features: Confirm agitator safely shuts down in case of anomalies or faults	FRS_PR_12	Agitator safely shuts down in case of anomalies			
OQ_18	Noise and Vibration: Measure and document noise and vibration levels	FRS_PR_13	Noise and vibration levels comply with standards			

No.	Instructions	Reference	Expected Results	Actual Result	Result (Pass/Fail)	Signature
OQ_19	Noise and Vibration: Confirm compliance of equipment with safety and environmental standards	FRS_PR_14	Equipment is compliant with safety and environmental standards			
OQ_20	Material Compatibility: Confirm material compatibility with various processed materials	FRS_PR_15	Agitator is compatible with various processed materials			
OQ_21	Material Compatibility: Verify if there are any material specific issues (Eg. Corrosion)	FRS_PR_16	No material specific issues are observed			
OQ_22	Operational Reliability: Verify continuous operation of agitator to assess long-term reliability	FRS_PR_17	Agitator demonstrates long-term reliability			
OQ_23	Operational Reliability: Monitor for signs of wear, overheating etc. during continuous operation	FRS_PR_18	No signs of wear, overheating or similar issues are observed			
OQ_24	Documentation: Verify thorough documentation of parameters and test results	FRS_PR_19	Thorough documentation of parameters and results			

No.	Instructions	Reference	Expected Results	Actual Result	Result (Pass/Fail)	Signature
OQ_25	Documentation: Compliance with regulatory and quality standards	FRS_PR_20	Equipment is compliant with regulatory and quality standards			

Operational Qualification Test Criteria 4

☐ Mark if not applicable

No.	Instructions	Reference	Expected Results	Actual Result	Specification Met?	Signature	Comments
OQ_26	Agitation Speed: Verify specified speed range within acceptable tolerance of +/- 5%	FRS_PR_01	Equipment speed is with acceptable tolerance of +/- 5%		Met Not Met		
OQ_27	Agitation Speed: Confirm that the agitator operates at consistent speed within tolerances.	FRS_PR_02	Equipment operates at consistent speed within tolerances.		Met Not Met		
OQ_28	Mixing direction: Confirm that the agitator startup & stop is smooth	FRS_PR_03	Smooth startup & stop functionality is observed		Met Not Met		
OQ_29	Mixing direction: Test the clockwise(CW) and counter clockwise(CCW) operation of agitator	FRS_PR_04	Agitator changes direction(CW and CCW) as required		Met Not Met		

No.	Instructions	Reference	Expected Results	Actual Result	Specification Met?	Signature	Comments
OQ_30	Mixing Intensity: Confirm smooth direction changes of agitator from CW and CCW	FRS_PR_05	Agitator direction change occurs smoothly, without jolts.		Met Not Met		
OQ_31	Mixing Intensity: Confirm that provides required mixing intensity ensuring effective mixing of components.	FRS_PR_06	Agitator provides required mixing intensity		Met Not Met		
OQ_32	Confirm that Mixing Intensity levels are adjustable within specified parameters	FRS_PR_07	Agitator Mixing Intensity is adjustable within parameters		Met Not Met		
OQ_33	Load Handling: Confirm that Agitator performance under varying load conditions within design capacity	FRS_PR_08	Agitator performs under varying load conditions		Met Not Met		
OQ_34	Temperature Control: Confirm that agitator maintains desired vessel temperature during mixing	FRS_PR_09	Agitator maintains desired vessel temperature		Met Not Met		
OQ_35	Temperature Control: Agitator responds to temperature set point changes	FRS_PR_10	Agitator prompts response to temperature set point changes		Met Not Met		

Operational Qualification Test Criteria 5

☐ Mark if not applicable

S.No	Test Description			Adheres to Specification (Yes/No)
OQ_36	Agitation Speed Parameter Verification:			
	S. No	Test description	Achieves (Yes/No)	
	1	Set 10 rpm as the target speed for agitator using the control knobs on the equipment control panel		
	2	Set 50 rpm as the target speed for agitator using the control knobs on the equipment control panel		
	3	Set 100 rpm as the target speed for agitator using the control knobs on the equipment control panel		
OQ_37.a	Pump Accuracy Verification (Delay Parameters):			
	S. No	Test description	Ref. Value	
	1	Delay Time (in milliseconds)	1-3 ms	
	2	Delay Volume (millilitres)	50-75 ml	
OQ_37.b	Pump Accuracy Verification (Mixing Parameters)			
	S. No	Test description	Ref. Value	
	1	Delay Time (in milliseconds)	1-3 ms	
	2	Delay Volume (millilitres)	50-75 ml	

6 SUMMARIZE OPERATIONAL QUALIFICATION TEST CRITERIA:

Description	Results
The equipment is operating in accordance with the manufacturer specifications on restoring power to the equipment.	☐ Yes ☐ No ☐ Exp.No. _____
Verified By:	Date:

1 INTRODUCTION

Agitators play a pivotal role in facilitating the homogeneous mixing of substances, a fundamental process in pharmaceutical manufacturing. OQ, in this context, serves as a comprehensive evaluation and verification process, encompassing various key parameters such as agitation speed, mixing direction, intensity, load handling capabilities, temperature control, safety features, noise and vibration levels, material compatibility, operational reliability, and documentation.

1.1 ABBREVIATIONS

OQ - Operational Qualification
SOP - Standard Operating Procedure
PFM - Revolutions per Minute
HVAC - Heating, Ventilation, and Air Conditioning
HMI - Human-Machine Interface
PLC - Programmable Logic Controller

2 PURPOSE

The purpose of OQ is to rigorously assess the agitator's ability to function within specified parameters, ensuring it meets not only process requirements but also stringent regulatory and safety standards. A successful OQ not only validates the agitator's performance but also enhances process consistency, product quality, and overall operational excellence in the chemical processing industry. The OQ Protocol section of this qualification package defines and validates that the Agitator will perform its operational functions repeatedly, reliably, and as expected throughout the life of the system. The OQ Protocol documentation includes confirmation reports of operational functionality tests used to verify that the Agitator will perform its operational functions repeatedly, reliably, and as expected throughout the life of the system. The OQ Protocol documentation includes quality inspection reports, and checklists of the Equipment System labeling and documentation. The OQ Protocol documentation also includes the Operating and Maintenance Manual.

3 SYSTEM DESCRIPTION

Equipment : Agitator
Function : Agitation/ Mixing of Liquid Samples
Year of manufacture : 2022-2023

(Equipment Id. _____)

4 SCOPE

This protocol applies to the operational qualification of Agitator

4.1 PROTOCOL ACCEPTANCE CRITERIA

- All required test steps have been performed and actual results meet the expected results.
- All corresponding attachments are completed.
- All tests are passed and/or exceptional conditions have been adequately resolved and approved.



Dynamic Agitator



Magnetic Agitator

5 PRE-REQUISITES

Prior to executing qualification, following condition must be fulfilled:

Sr. No.	Pre-requisites	Availability (Yes / No / NA)	Document no. & location
1.	SAT protocol of equipment is pre-approved.		
2.	Equipment is installed and tagged.		
3.	Vessel is empty, clean, and not pressurized.		
4.	All utilities are available and connected to system.		
5.	All utilities are available as per Pre SAT-Check list		

5 OPERATIONAL QUALIFICATION SECTION:

Operational Qualification Test Criteria 1

☐ Mark if not applicable

No.	Instructions	Reference	Expected Results	Actual Result	Result (Pass/Fail)	Signature
OQ_1	Face Velocity Test: Turn on main power, turn on control box power, set speed to deliver 72-108 fpm. Measure airflow using calibrated device	FRS_FS_01	Equipment powers up, fans start and delivers 72-108 fpm.			
OQ_2	Static Particulate Test: Measure room particulates at the work surface using a calibrated meter. Measure in grid within tent work area (minimum 8 sample points)	FRS_FS_02	Particulate count less than 10,000 - 0.5-micron particles.			
OQ_3	Pressurization: Measure the differential pressure from the inside of the tent to the outside using a calibrated meter	FRS_FS_03	Differential pressure is positive from inside of tent to outside			
OQ_4	Room Change Rate: Calculate tent room change rates. (512 cu.ft. room volume; design airflow 700 cfm per filter)	FRS_MBS_01	Verify 60 changes per hour			
OQ_5	Manually operate the control switches listed in the applicable attachment	FRS_MBS_02	All operator controls and indicators tested shall conform to the expected response			

No.	Instructions	Reference	Expected Results	Actual Result	Result (Pass/Fail)	Signature
OQ_6	Confirmation of Receipt of all Required Parts: 1. Review manufacturing drawing, bill-of-materials, and sales order 2. Confirm that all loose plumbing and/or electrical components were shipped	FRS_MBS_03	All plumbing and/or electrical components were shipped			
OQ_7	Check whether the Equipment maintains temperature of -70°C for 7 days	FRS_DS_01	Maintains temperature over 7 days			
OQ_8	Check whether temperature LED functions correctly	FRS_DS_02	LED temperature display functions correctly			
OQ_9	Confirmation of Components are Leak Tight after Connection to LN2 Supply: 1. Review safe handling of liquid nitrogen guidelines 2. Hook up liquid nitrogen supply to T-W Equipment per Operating Manual 3. Open fill valve on liquid nitrogen supply 4. Inspect component connections for leaks using leak test procedures 5. Tighten any leaks, if required	FRS_DS_03	Components are leak tight after connection to LN2 supply			

No.	Instructions	Reference	Expected Results	Actual Result	Result (Pass/Fail)	Signature
OQ_10	Confirmation of High Temperature Alarm Set Point: Confirm whether Alarm rings if Temperature rises more than the optimum level	FRS_DS_04	Alarm Rings once temperature rises more than the optimum level			
OQ_11	Confirmation of Low Temperature Alarm Set Point: Confirm whether Alarm rings if Temperature drops below the optimum level	FRS_DS_05	Alarm Rings once temperature drops below the optimum level			

Operational Qualification Test Criteria 2

☐ Mark if not applicable

No.	Instructions	Reference	Expected Results	Actual Result	Result (Pass/Fail)	Signature
IQ_12	Verification should be done to confirm whether erosion resistant coatings based on the design of the turbine or pump and the severity of erosion are used	FRS_FS_04	Verification for erosion resistant coatings is successful			
IQ_13	Equipment verification done to ensure that the machine has maintained GMP conditions at all times with no possibility for product mix or for rejected product to be released.	FRS_FS_05	Equipment Verification for GMP maintenance is successful			

No.	Instructions	Reference	Expected Results	Actual Result	Result (Pass/Fail)	Signature
IQ_14	Verification is done to confirm environment setting is as described in the ICD (Interface Control Documents)	FRS_IS_01	Environment setting as per ICD (Interface Control Documents) is verified successfully			
IQ_15	Verification to be done to confirm whether equipment complies with the ISS System Services interface	FRS_IS_02	Equipment compilation with the ISS System Services interface is verified successfully			

Operational Qualification Test Criteria 3

☐ Mark if not applicable

No.	Instructions	Reference	Expected Results	Actual Result	Result (Pass/Fail)	Signature
OQ_16	Safety Features: Confirm Agitator includes functional safety interlocks and emergency stop	FRS_PR_11	Functional safety interlocks and emergency stop operate			
OQ_17	Safety Features: Confirm agitator safely shuts down in case of anomalies or faults	FRS_PR_12	Agitator safely shuts down in case of anomalies			
OQ_18	Noise and Vibration: Measure and document noise and vibration levels	FRS_PR_13	Noise and vibration levels comply with standards			

No.	Instructions	Reference	Expected Results	Actual Result	Result (Pass/Fail)	Signature
OQ_19	Noise and Vibration: Confirm compliance of equipment with safety and environmental standards	FRS_PR_14	Equipment is compliant with safety and environmental standards			
OQ_20	Material Compatibility: Confirm material compatibility with various processed materials	FRS_PR_15	Agitator is compatible with various processed materials			
OQ_21	Material Compatibility: Verify if there are any material specific issues (Eg. Corrosion)	FRS_PR_16	No material specific issues are observed			
OQ_22	Operational Reliability: Verify continuous operation of agitator to assess long-term reliability	FRS_PR_17	Agitator demonstrates long-term reliability			
OQ_23	Operational Reliability: Monitor for signs of wear, overheating etc. during continuous operation	FRS_PR_18	No signs of wear, overheating or similar issues are observed			
OQ_24	Documentation: Verify thorough documentation of parameters and test results	FRS_PR_19	Thorough documentation of parameters and results			

No.	Instructions	Reference	Expected Results	Actual Result	Result (Pass/Fail)	Signature
OQ_25	Documentation: Compliance with regulatory and quality standards	FRS_PR_20	Equipment is compliant with regulatory and quality standards			

Operational Qualification Test Criteria 4

☐ Mark if not applicable

No.	Instructions	Reference	Expected Results	Actual Result	Specification Met?	Signature	Comments
OQ_26	Agitation Speed: Verify specified speed range within acceptable tolerance of +/- 5%	FRS_PR_01	Equipment speed is with acceptable tolerance of +/- 5%		Met Not Met		
OQ_27	Agitation Speed: Confirm that the agitator operates at consistent speed within tolerances.	FRS_PR_02	Equipment operates at consistent speed within tolerances.		Met Not Met		
OQ_28	Mixing direction: Confirm that the agitator startup & stop is smooth	FRS_PR_03	Smooth startup & stop functionality is observed		Met Not Met		
OQ_29	Mixing direction: Test the clockwise(CW) and counter clockwise(CCW) operation of agitator	FRS_PR_04	Agitator changes direction(CW and CCW) as required		Met Not Met		

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OQ_30	Mixing Intensity: Confirm smooth direction changes of agitator from CW and CCW	FRS_PR_05	Agitator direction change occurs smoothly, without jolts.		Met Not Met		
OQ_31	Mixing Intensity: Confirm that provides required mixing intensity ensuring effective mixing of components.	FRS_PR_06	Agitator provides required mixing intensity		Met Not Met		
OQ_32	Confirm that Mixing Intensity levels are adjustable within specified parameters	FRS_PR_07	Agitator Mixing Intensity is adjustable within parameters		Met Not Met		
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OQ_35	Temperature Control: Agitator responds to temperature set point changes	FRS_PR_10	Agitator prompts response to temperature set point changes		Met Not Met		

Operational Qualification Test Criteria 5

☐ Mark if not applicable

S.No	Test Description			Adheres to Specification (Yes/No)
OQ_36	Agitation Speed Parameter Verification:			
	S. No	Test description	Achieves (Yes/No)	
	1	Set 10 rpm as the target speed for agitator using the control knobs on the equipment control panel		
	2	Set 50 rpm as the target speed for agitator using the control knobs on the equipment control panel		
	3	Set 100 rpm as the target speed for agitator using the control knobs on the equipment control panel		
OQ_37.a	Pump Accuracy Verification (Delay Parameters):			
	S. No	Test description	Ref. Value	
	1	Delay Time (in milliseconds)	1-3 ms	
	2	Delay Volume (millilitres)	50-75 ml	
OQ_37.b	Pump Accuracy Verification (Mixing Parameters)			
	S. No	Test description	Ref. Value	
	1	Delay Time (in milliseconds)	1-3 ms	
	2	Delay Volume (millilitres)	50-75 ml	

6 SUMMARIZE OPERATIONAL QUALIFICATION TEST CRITERIA:

Description	Results
The equipment is operating in accordance with the manufacturer specifications on restoring power to the equipment.	☐ Yes ☐ No ☐ Exp.No. _____
Verified By:	Date:

1 INTRODUCTION

Agitators play a pivotal role in facilitating the homogeneous mixing of substances, a fundamental process in pharmaceutical manufacturing. OQ, in this context, serves as a comprehensive evaluation and verification process, encompassing various key parameters such as agitation speed, mixing direction, intensity, load handling capabilities, temperature control, safety features, noise and vibration levels, material compatibility, operational reliability, and documentation.

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3 SYSTEM DESCRIPTION

Equipment : Agitator
Function : Agitation/ Mixing of Liquid Samples
Year of manufacture : 2022-2023

(Equipment Id. _____)

4 SCOPE

This protocol applies to the operational qualification of Agitator

4.1 PROTOCOL ACCEPTANCE CRITERIA

- All required test steps have been performed and actual results meet the expected results.
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Dynamic Agitator



Magnetic Agitator

5 PRE-REQUISITES

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5 OPERATIONAL QUALIFICATION SECTION:

Operational Qualification Test Criteria 1

☐ Mark if not applicable

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☐ Mark if not applicable

No.	Instructions	Reference	Expected Results	Actual Result	Result (Pass/Fail)	Signature
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Operational Qualification Test Criteria 3

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No.	Instructions	Reference	Expected Results	Actual Result	Result (Pass/Fail)	Signature
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OQ_37.a	Pump Accuracy Verification (Delay Parameters):			
	S. No	Test description	Ref. Value	
	1	Delay Time (in milliseconds)	1-3 ms	
	2	Delay Volume (millilitres)	50-75 ml	
OQ_37.b	Pump Accuracy Verification (Mixing Parameters)			
	S. No	Test description	Ref. Value	
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	2	Delay Volume (millilitres)	50-75 ml	

6 SUMMARIZE OPERATIONAL QUALIFICATION TEST CRITERIA:

Description	Results
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3 SYSTEM DESCRIPTION

Equipment : Agitator
Function : Agitation/ Mixing of Liquid Samples
Year of manufacture : 2022-2023

(Equipment Id. _____)

4 SCOPE

This protocol applies to the operational qualification of Agitator

4.1 PROTOCOL ACCEPTANCE CRITERIA

- All required test steps have been performed and actual results meet the expected results.
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Dynamic Agitator



Magnetic Agitator

5 PRE-REQUISITES

Prior to executing qualification, following condition must be fulfilled:

Sr. No.	Pre-requisites	Availability (Yes / No / NA)	Document no. & location
1.	SAT protocol of equipment is pre-approved.		
2.	Equipment is installed and tagged.		
3.	Vessel is empty, clean, and not pressurized.		
4.	All utilities are available and connected to system.		
5.	All utilities are available as per Pre SAT-Check list		

5 OPERATIONAL QUALIFICATION SECTION:

Operational Qualification Test Criteria 1

☐ Mark if not applicable

No.	Instructions	Reference	Expected Results	Actual Result	Result (Pass/Fail)	Signature
OQ_1	Face Velocity Test: Turn on main power, turn on control box power, set speed to deliver 72-108 fpm. Measure airflow using calibrated device	FRS_FS_01	Equipment powers up, fans start and delivers 72-108 fpm.			
OQ_2	Static Particulate Test: Measure room particulates at the work surface using a calibrated meter. Measure in grid within tent work area (minimum 8 sample points)	FRS_FS_02	Particulate count less than 10,000 - 0.5-micron particles.			
OQ_3	Pressurization: Measure the differential pressure from the inside of the tent to the outside using a calibrated meter	FRS_FS_03	Differential pressure is positive from inside of tent to outside			
OQ_4	Room Change Rate: Calculate tent room change rates. (512 cu.ft. room volume; design airflow 700 cfm per filter)	FRS_MBS_01	Verify 60 changes per hour			
OQ_5	Manually operate the control switches listed in the applicable attachment	FRS_MBS_02	All operator controls and indicators tested shall conform to the expected response			